

# Excluding the Bounded Negative Part

*Lean-checked rigidity for Erdős Problem #243*

WILL COOK

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## ABSTRACT

Sylvester's sequence  $2, 3, 7, 43, 1807, \dots$  satisfies  $a_n = a_{n-1}^2 - a_{n-1} + 1$  and has reciprocal sum 1. Erdős asked whether a rapidly growing integer sequence with rational reciprocal sum must satisfy that recurrence eventually. Clearing denominators gives integer states  $D_{n+1} = a_n D_n$  and  $C_{n+1} = a_n C_n - D_n$ , and centring at the Sylvester tail gives  $E_n = D_n - (a_n - 1)C_n$ , with  $C_{n+1} = C_n - E_n$ ; the question becomes whether  $E$  can avoid vanishing. When  $E$  is nonnegative a one-line descent settles it. This note is about the negative side.

We formalise in Lean three increasingly general exclusions of a negative  $E$ . It cannot be constantly  $-m$ , at any magnitude and any scale: every multiplier would have to share a prime with the fixed  $m$ , and once a prime divides one multiplier every later multiplier is 1 modulo it, so a fixed integer would need infinitely many distinct prime divisors. It cannot be periodic with positive drift, by the same pigeonhole after a strong induction that removes common scale. And, under eventual strict centring and normalised vanishing, it cannot have a bounded negative part: a bounded negative part makes  $C$  rise by a bounded amount, normalised vanishing makes  $C$  tend to infinity, the tail gcd stabilises so the orbit may be taken reduced, and in a reduced tail the multipliers are pairwise coprime and permanently avoided by the numerator, which a Chinese-remainder block of consecutive multiples forbids. That last step is a statement about integer sequences alone: a sequence tending to infinity with upward steps bounded by  $B$  cannot permanently avoid infinitely many pairwise coprime moduli. Together with an absorption lemma making  $E_n = 0$  propagate, this yields: under exact dynamics, eventual strict centring, an eventually bounded negative part, and normalised vanishing,  $E$  vanishes eventually, and the sequence satisfies the Sylvester recurrence eventually.

Problem #243 is open and this note does not close it. The last theorem is conditional, and two of its hypotheses are analytic inputs that we assume rather than derive. What the results remove is the bounded case and, more generally, the finite normalised negative-mass case. Under normalised vanishing, summability of  $\sum_n (-E_n)_+ / C_n$  makes  $C_n$  bounded and hence  $E_n$  eventually zero. Thus any surviving exact state orbit must have negative excursions unbounded along a cofinal set and divergent normalised negative mass.

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# 1 Introduction

Sylvester's sequence 2, 3, 7, 43, 1807, ... is generated by  $a_n = a_{n-1}^2 - a_{n-1} + 1$  and has reciprocal sum 1; the recurrence is exactly the statement that the tail beyond  $a_{n-1}$  equals  $1/(a_{n-1} - 1)$ . Erdős Problem #243 asserts that this is the only way a sequence of that growth can have a rational reciprocal sum:

**Problem 1.1** (Erdős #243). Let  $1 \leq a_1 < a_2 < \dots$  be a sequence of integers with

$$\lim_{n \rightarrow \infty} \frac{a_n}{a_{n-1}^2} = 1 \quad \text{and} \quad \sum \frac{1}{a_n} \in \mathbb{Q}.$$

Then  $a_n = a_{n-1}^2 - a_{n-1} + 1$  for all sufficiently large  $n$ .

See [3, p. 64] and [4, p. 105]; numbering and status follow Bloom's catalogue [8]. The problem is open. Below we index the same recurrence as  $a_{n+1} = a_n^2 - a_n + 1$ , which is the shift the formal sources use; the two forms are the same statement. Write  $\text{syl}(a) = a^2 - a + 1$ , the [Sylvester successor](#).

**Relation to prior work.** Two results bound what is new here, and both are stronger than anything below. Erdős and Straus proved that if  $\lim a_n/a_{n-1}^2 = 1$ , the reciprocal sum is rational, and  $a_n$  does not satisfy the recurrence, then

$$\limsup_{n \rightarrow \infty} \frac{[a_1, \dots, a_n]}{a_{n+1}} \left( \frac{a_n^2}{a_{n+1}} - 1 \right) > 0,$$

where  $[a_1, \dots, a_n]$  is the least common multiple [1]. Duverney proved a conditional form of the problem itself: if  $\sum_{n \geq 0} (a_{n+1}/a_n^2 - 1)$  converges, then the reciprocal sum is rational if and only if the recurrence holds for all large  $n$  [2]. Neither is formalised here, and the results below do not recover either of them.

The growth hypothesis in Problem 1.1 is calibrated by two facts about Sylvester's sequence. It satisfies  $a_n \approx c_0^{2^n}$  with  $c_0 = 1.2640847 \dots$  (Graham, Knuth and Patashnik [6, p. 109]; Vardi [7]), and shifting it further produces sequences with  $a_n \approx C^{2^n}$  for arbitrarily large  $C$  whose reciprocals still sum to a rational number. The classical sufficient condition for irrationality,  $\lim_n a_n^{1/2^n} = \infty$ , is therefore sharp. That condition is folklore [1], the sharpness observation is due to Erdős (1975), and both are taken here from the introduction of Kovač and Tao [5]. A sequence with  $a_n/a_{n-1}^2 \rightarrow 1$  has  $a_n^{1/2^n}$  convergent, so that criterion says nothing about the sequences of Problem 1.1, and rationality is genuinely possible there. Sylvester's sequence is A000058 in the OEIS and its shifts A129871. For the recent literature on irrationality of Ahmes series we refer to Kovač and Tao [5], who resolve several problems of Erdős and Graham drawn from the same two sources cited above, and whose introduction gives a sample of the intermediate work, including Sándor (1984) and Badea (1987). They do not treat Problem #243; the rigidity conclusion asked for there is not among their results.

The statement of Problem #243 has also been formalised before, as a conjecture with an unfilled proof, in the *Formal Conjectures* collection [9]; that is a statement of the question, and the development below is independent of it. What is machine-checked below

concerns the state system rather than the analytic problem: the exclusion of constant and periodic negative magnitudes, the bounded-rise barrier of Theorem 7.2, and the two conditional endpoints (Theorems 8.1 and 9.1), each stated with the exact hypotheses that separate it from Problem #243. No claim of priority is made for anything below.

The hypothesis is asymptotic and the conclusion is exact, so the argument must convert an analytic rate into an integer obstruction. The conversion used here is the classical one: clear denominators along the sequence, so that rationality makes a tail integral, and then centre that integer at the value it would take on Sylvester's sequence. What remains is a single integer error  $E$ , and the whole question is whether  $E$  can avoid vanishing.

*Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no sorry, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Taken together the exclusions constrain the shape of any counterexample. By Theorem 3.4 a counterexample has  $E_n \neq 0$  for every large  $n$ . By Theorem 4.1 it cannot have  $E$  eventually nonnegative. By Theorems 5.1 and 6.1 its negative magnitudes are neither eventually constant nor eventually periodic. And by Theorems 8.1 and 9.1, if the two analytic hypotheses of Section 8 hold, the negative part is neither bounded nor of finite normalised mass. What survives is stated in Section 10.

## Structure

Section 2 sets up the state system, and Section 3 proves the defect identity, from which eventual vanishing of  $E$  forces the Sylvester recurrence and a single vanishing error propagates. Section 4 disposes of the case  $E \geq 0$  in three lines. The rest of the note is the case  $E < 0$ , in four widening steps: constant magnitude (Section 5), periodic magnitude (Section 6), bounded magnitude via a coprimality barrier (Sections 7 and 8), finite normalised negative mass (Section 9), and what is left (Section 10). These sections contain the new content; a reader who wants only the strongest statements should read Sections 8 and 9, and then Section 10. Appendix A is a guide to the formal sources, and Appendix B records an exact residue reduction that an earlier finite search used. Linked phrases open the corresponding Lean declaration at the pinned source revision 580ae3a8bcce.

**Keywords.** irrationality; Ahmes series; Sylvester's sequence; unit fractions; Lean 4. **MSC 2020.** 11J72 (primary); 11B37, 11D68, 68V20 (secondary).

## 2 The centred integer state

For  $a, D, C \in \mathbb{Z}$  define

$$\text{den}(a, D) = aD, \quad \text{tail}(a, D, C) = aC - D, \quad \text{ctr}(a, D, C) = D - (a - 1)C,$$

the [denominator update](#), the [tail update](#), and the [centred state](#). Given  $(a_n)$  we write  $D_{n+1} = \text{den}(a_n, D_n)$ ,  $C_{n+1} = \text{tail}(a_n, D_n, C_n)$  and  $E_n = \text{ctr}(a_n, D_n, C_n)$ .

| Statement                                     | Status                                           | Treatment here                                                           |
|-----------------------------------------------|--------------------------------------------------|--------------------------------------------------------------------------|
| Problem #243 as stated                        | Open                                             | Not proved.                                                              |
| Defect identity                               | Proved here                                      | Theorem 3.1, an identity in $\mathbb{Z}[a, a', D, C]$ .                  |
| Eventual vanishing forces the recurrence      | Proved here                                      | Theorem 3.3.                                                             |
| Zero is absorbing                             | Proved here                                      | Theorem 3.4, under strict centring.                                      |
| Nonnegative $E$                               | Proved here                                      | Theorem 4.1, by descent.                                                 |
| Constantly negative $E$                       | Excluded here                                    | Theorem 5.1, at any magnitude and scale.                                 |
| Periodic negative magnitude                   | Excluded here                                    | Theorem 6.1, in the regime $e_n < a_n$ and with positive drift $M > 0$ . |
| Coprime-modulus barrier                       | Proved here                                      | Theorem 7.2.                                                             |
| Bounded negative part                         | Excluded here; conditional                       | Theorem 8.1; see its hypotheses.                                         |
| Finite normalised negative mass               | Excluded here; conditional                       | Theorem 9.1, Section 9; normalised vanishing remains an input.           |
| Factorial residue reduction                   | Proved here; superseded for its original purpose | Theorem B.1, Appendix B.                                                 |
| Strict centring, normalised vanishing         | Assumed, not derived                             | Section 8.                                                               |
| Unbounded, divergent-mass negative excursions | Open                                             | Section 10; the surviving obstruction.                                   |
| Erdős–Straus criterion; Duverney              | Proved elsewhere                                 | Cited, not formalised; both stronger than anything here.                 |
| State system equals reciprocal tails          | Not formalised                                   | Section 2.                                                               |

**Proposition 2.1** (update law).  $\text{tail}(a, D, C) = C - \text{ctr}(a, D, C)$ ; that is,  $C_{n+1} = C_n - E_n$ .

*Proof.*  $C - (D - (a - 1)C) = aC - D$ . □

Formalised as the [update law](#). The state therefore moves by exactly the error at each step, which is what makes the sign of  $E$  decisive: where  $E$  is positive the state falls, where  $E$  is negative it rises, and where  $E$  vanishes it is stationary.

**The intended reading, which is not formalised.** Let  $T_n = \sum_{k \geq n} 1/a_k$  and suppose  $T_0 \in \mathbb{Q}$ . Put  $D_0$  equal to a common denominator and  $D_n = D_0 a_0 \cdots a_{n-1}$ , and set  $C_n = D_n T_n$ . Then  $C_n \in \mathbb{Z}$  for every  $n$ , and from  $T_n = 1/a_n + T_{n+1}$  one gets  $C_{n+1} = a_n C_n - D_n$ , which is the tail update. On Sylvester’s sequence  $T_n = 1/(a_n - 1)$  exactly, so  $D_n = (a_n - 1)C_n$  and  $E_n = 0$ : the centred state measures deviation from the Sylvester tail identity, and it vanishes identically on the Sylvester orbit.

None of this paragraph is formalised. The Lean module contains no series, no rationality hypothesis, and no growth hypothesis; it proves identities and implications

about the integer system above, and about its natural-number realisation, for which the bridging identities are the [tail realisation](#), the [denominator realisation](#), and the [signed update law](#). A reader should treat the identification with reciprocal tails as motivation for the definitions and not as a checked step.

**Proposition 2.2** (scale equivariance). *For every  $s, a, D, C \in \mathbb{Z}$ ,*

$$\begin{aligned} \text{den}(a, sD) &= s \text{ den}(a, D), & \text{tail}(a, sD, sC) &= s \text{ tail}(a, D, C), \\ \text{ctr}(a, sD, sC) &= s \text{ ctr}(a, D, C). \end{aligned}$$

Formalised as the [denominator equivariance](#), the [tail equivariance](#), and the [centred equivariance](#). The system depends only on the ray through  $(D, C)$ . This is used twice below: once to normalise the finite search of Appendix B, and once, in Theorem 6.1, as the induction step that removes common scale.

### 3 The defect identity and its two consequences

Write  $\text{def}(a, a') = a' - \text{syl}(a)$  for the deviation of the next term from the Sylvester successor, the [Sylvester defect](#).

**Theorem 3.1** (defect identity). *For all  $a, a', D, C \in \mathbb{Z}$ ,*

$$\text{def}(a, a') \cdot \text{tail}(a, D, C) = a^2 \text{ ctr}(a, D, C) - \text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)),$$

that is,  $\Delta_n C_{n+1} = a_n^2 E_n - E_{n+1}$ .

*Proof.* A direct expansion: both sides equal  $a' a C - a' D - a^3 C + a^2 D + a^2 C - a D - a C + D$ .  $\square$

Formalised as the [defect identity](#). It is a polynomial identity in four indeterminates, with no hypothesis of any kind, and it carries the whole argument twice over: once for rigidity, and once for absorption.

**Theorem 3.2** (two vanishing errors force the step). *Let  $a, a', D, C \in \mathbb{Z}$  with  $\text{tail}(a, D, C) \neq 0$ . If  $\text{ctr}(a, D, C) = 0$  and  $\text{ctr}(a', \text{den}(a, D), \text{tail}(a, D, C)) = 0$ , then  $a' = \text{syl}(a)$ .*

*Proof.* By Theorem 3.1 the product  $\text{def}(a, a') \cdot \text{tail}(a, D, C)$  is zero, and the second factor is nonzero by hypothesis.  $\square$

Formalised as the [local rigidity step](#). The hypothesis  $C_{n+1} \neq 0$  is not removable: at  $C_{n+1} = 0$  the identity gives no information about  $a'$ .

**Theorem 3.3** (sequence form). *Let  $a, D, C : \mathbb{N} \rightarrow \mathbb{Z}$  satisfy  $D_{n+1} = \text{den}(a_n, D_n)$  and  $C_{n+1} = \text{tail}(a_n, D_n, C_n)$ . If  $E_n = 0$  for all sufficiently large  $n$  and  $C_{n+1} \neq 0$  for all sufficiently large  $n$ , then  $a_{n+1} = a_n^2 - a_n + 1$  for all sufficiently large  $n$ .*

Formalised as the [eventual Sylvester recurrence](#). The proof is routine: take the larger of the two thresholds and apply Theorem 3.2 at each later index. Every later section is directed at the hypothesis of this theorem, that is, at forcing  $E$  to vanish eventually.

The defect identity has a second consequence, which is what makes a single vanishing error worth having.

**Theorem 3.4** (zero is absorbing). *Let  $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$  be an exact natural orbit, so  $C_{n+1} + D_n = a_n C_n$  and  $D_{n+1} = a_n D_n$ , and let  $E_n = \text{ctr}(a_n, D_n, C_n)$ . Suppose the centring is strict,  $|E_n| < C_n$  for every  $n$ . If  $E_n = 0$  then  $E_{n+1} = 0$ .*

*Proof.* Putting  $E_n = 0$  in Theorem 3.1 gives  $\Delta_n C_{n+1} = -E_{n+1}$ , so  $C_{n+1}$  divides  $E_{n+1}$ . A multiple of  $C_{n+1}$  of absolute value smaller than  $C_{n+1}$  is zero, and  $|E_{n+1}| < C_{n+1}$  by hypothesis.  $\square$

Formalised as the [absorption of a vanishing centred state](#), with the companion [contra-positive along a tail](#): if zero is absorbing beyond some index and  $E$  does not vanish eventually, then  $E$  is nowhere zero beyond that index.

Absorption changes the shape of the problem. Without it, one would have to exclude an  $E$  that dips to zero and recovers. With it, either  $E$  hits zero once and is finished, or it is never zero at all; the later sections may therefore assume that  $E_n \neq 0$  for every  $n$ , and in the negative case that  $E_n < 0$  throughout, without loss.

## 4 Descent when the error is nonnegative

**Theorem 4.1** (descent). *Let  $C, E : \mathbb{N} \rightarrow \mathbb{N}$  satisfy  $C_{n+1} + E_n = C_n$  for every  $n$ . Then  $E_n = 0$  for all sufficiently large  $n$ .*

*Proof.* A one-line descent. The relation forces  $C_{n+1} \leq C_n$ , so  $C$  is a nonincreasing sequence of natural numbers and is eventually equal to its minimum; beyond that index  $C_{n+1} = C_n$ , and the relation gives  $E_n = 0$ .  $\square$

Formalised as the [stabilisation of the nonnegative error](#), on the [eventual constancy of a nonincreasing sequence](#).

The hypothesis is exactly one-sidedness:  $C$  and  $E$  take values in  $\mathbb{N}$ . If  $E_n < 0$  at infinitely many indices then  $C$  rises there, nothing descends, and the argument is unavailable. That is the whole difficulty, and the rest of this note is directed at it.

## 5 Excluding a constant negative magnitude

Suppose  $E_n = -m$  for a fixed  $m > 0$  and every  $n$ . By Proposition 2.1 the state then climbs by exactly  $m$  each step, so  $C_n = c + nm$  with  $c = C_0$ , and the definition of the centred state becomes a *shape equation*

$$D_n + m = (a_n - 1)(c + nm). \quad (5.1)$$

Together with  $D_{n+1} = a_n D_n$  this is a closed system in  $(a, D)$ , and it has no solutions.

**Theorem 5.1** (no constant negative magnitude). *There is no pair of sequences  $a, D : \mathbb{N} \rightarrow \mathbb{N}$  with  $a_n \geq 2$  for all  $n$  satisfying  $D_{n+1} = a_n D_n$  and (5.1), for any  $m > 0$  and any  $c$ . The same holds if the shape only begins at some index.*

*Proof of the scale-primitive case.* Assume first  $\gcd(c, m) = 1$ . Two facts about the multipliers do all the work, and the difficulty lies in the fact that they concern the same finite set

of primes from opposite directions: the first makes every multiplier meet that set, the second lets each prime of the set meet at most one multiplier.

Every  $a_j$  shares a prime with  $m$ . Suppose  $\gcd(a_j, m) = 1$ . Then  $m$  is invertible modulo  $a_j$ , so some  $n$  has  $c + nm \equiv 0$ , that is  $a_j \mid C_n$ ; and  $a_j \mid D_n$  for every  $n > j$ , since  $D$  is multiplicative with  $a_j$  among its factors. Choosing such an  $n$  beyond  $j$  and reading (5.1) modulo  $a_j$  gives  $a_j \mid m$ , contradicting  $\gcd(a_j, m) = 1$  and  $a_j \geq 2$ .

A prime divisor of  $m$  occurs in at most one  $a_j$ . Suppose  $p \mid m$  and  $p \mid a_j$ . For  $n > j$  we have  $p \mid D_n$ , so (5.1) gives  $(a_n - 1)C_n \equiv 0 \pmod{p}$ . Now  $C_n = c + nm \equiv c$ , and  $p \nmid c$  because  $p \mid m$  and  $\gcd(c, m) = 1$ ; hence  $p \mid a_n - 1$ , so  $p \nmid a_n$ . Thus  $p$  cannot divide any later multiplier.

The two facts are incompatible. Each of the infinitely many multipliers needs a prime divisor of  $m$ , and each prime divisor of  $m$  serves at most one multiplier, while  $m$  has only finitely many prime divisors.  $\square$

The scale-primitive case is the [scale-primitive exclusion](#). The special case  $m = c = 1$ , where  $m$  has no prime divisors at all and the first fact is immediately contradictory, is recorded separately as the [normalised exclusion](#).

*Removing the scale.* For general  $c$ , put  $g = \gcd(c, m)$ . Equation (5.1) shows  $g \mid D_n$  for every  $n$ , so dividing  $D$ ,  $c$  and  $m$  by  $g$  leaves a system of the same shape with coprime data, which the previous case excludes.  $\square$

Formalised as the [exclusion at every scale](#), and the eventual form, obtained by shifting the orbit to the first constant index, as the [eventual exclusion](#).

Theorem 5.1 excludes a centred state that is eventually constant and negative, at every magnitude and every scale. It says nothing about a negative state whose magnitude changes, and the argument genuinely uses constancy: the finiteness of the set of prime divisors of  $m$  is what the pigeonhole uses, and a varying magnitude presents a fresh set at each index.

## 6 Excluding a periodic negative magnitude

The next case allows the magnitude to vary, provided it repeats. Write  $e_n = -E_n > 0$  for the magnitude, so  $C_{n+1} = C_n + e_n$ , and suppose  $e$  has period  $h$  and  $C$  gains a drift  $M > 0$  over one period.

Three lemmas replace the two facts of Section 5. The first is that multipliers persist: every  $a_j$  divides every strictly later denominator state, the [persistence of a multiplier](#). The second is a lock: a prime already dividing  $D$  and also dividing the current multiplier is forced into the next tail state, the [one-step lock](#), and once present in both  $D$  and  $C$  it stays in every later  $D$ ,  $C$  and magnitude, the [persistence of a lock](#). The third transports a lock backwards through the period: if a divisor of the drift  $M$  occurs in two multipliers, it is a common divisor of  $C_0$  and of every magnitude, the [repeated-divisor transport](#), using that both the period relation and the drift iterate over any number of periods, the [iterated period](#) and the [iterated drift](#).

**Theorem 6.1** (no periodic negative magnitude). *Let  $a, D, C, e : \mathbb{N} \rightarrow \mathbb{N}$  with  $a_n \geq 2$ ,  $e_n > 0$  and  $e_n < a_n$  for every  $n$ , satisfying*

$$D_{n+1} = a_n D_n, \quad C_{n+1} = C_n + e_n, \quad D_n + e_n = (a_n - 1)C_n,$$

*and suppose  $e_{n+h} = e_n$  and  $C_{n+h} = C_n + M$  for some  $h > 0$  and  $M > 0$ . This is impossible.*

*Proof.* Strong induction on the drift  $M$ . The difficulty lies in the fact that a repeated prime divisor of  $M$  need not contradict anything directly. It instead forces a common scale on the whole orbit, which can be divided out, and the induction runs on the drift that the division reduces.

Suppose first that no prime divisor of  $M$  is a common divisor of  $C_0$  and of every magnitude. Repeated-divisor transport then applies in the contrapositive: a prime divisor of  $M$  occurring in two of the multipliers would be exactly such a common divisor, so each prime divisor of  $M$  occurs in at most one multiplier. Against this, the argument of Section 5 run phase by phase (along a fixed residue class modulo  $h$  the magnitude is constant and  $C$  advances by  $M$  at each period, so the multipliers of that phase meet an arithmetic progression exactly as in Section 5) makes every multiplier share a prime with  $M$ . Since  $M$  has only finitely many prime divisors and there are infinitely many multipliers, pigeonhole gives a contradiction.

Otherwise some prime  $p$  divides  $M$ , divides  $C_0$ , and divides every magnitude. Then  $p \mid C_n$  for every  $n$ , since  $C_{n+1} = C_n + e_n$  and both summands on the right are divisible by  $p$ , and the shape equation  $D_n + e_n = (a_n - 1)C_n$  then gives  $p \mid D_n$  for every  $n$ . Divide  $D$ ,  $C$  and  $e$  by  $p$ . The multipliers are untouched, so  $a_n \geq 2$  and  $e_n < a_n$  persist and the magnitudes stay positive; the three recurrences are homogeneous in  $(D, C, e)$  and so survive; the period is still  $h$ ; and the drift becomes  $M/p < M$ . The inductive hypothesis applies.  $\square$

The phase-primitive case is the [phase-primitive exclusion](#), the induction is the [exclusion at every scale](#), and the eventual form is the [eventual exclusion](#).

The bound  $e_n < a_n$  is a genuine restriction and not a normalisation. It is the range in which a multiplier cannot divide its own phase magnitude, which is what the lock argument needs. It holds in the intended reading, where  $e_n$  is a centred representative and  $a_n$  grows quadratically, but it is assumed here and not derived. The drift hypothesis  $M > 0$  is also used: a periodic magnitude with zero drift would be identically zero, which is the case already settled by Section 4.

## 7 Bounded rise cannot avoid coprime moduli

Periodicity is still a strong assumption. Removing it needs a different kind of obstruction, and the one used here is a counting argument about where an integer sequence with bounded upward steps must land.

**Lemma 7.1** (shifted blocks of consecutive multiples). *Let  $m_0, \dots, m_{B-1}$  be pairwise coprime and at least 2. For every bound there is a  $t$  beyond it with  $m_i \mid t + i$  for each  $i < B$ .*

*Proof.* Standard Chinese remaindering: solve  $t \equiv -i \pmod{m_i}$  simultaneously, then add multiples of  $\prod_i m_i$  to pass the bound.  $\square$

Formalised as the [shifted consecutive multiples](#).

**Theorem 7.2** (bounded rise cannot avoid fresh moduli). *Let  $u : \mathbb{N} \rightarrow \mathbb{N}$  tend to infinity with  $u_{n+1} \leq u_n + B$  for a fixed  $B > 0$ . Then  $u$  cannot permanently avoid infinitely many fresh pairwise coprime moduli: there is no family of pairwise coprime  $m_i \geq 2$ , one for each index, such that  $m_i \nmid u_t$  whenever  $i < t$ .*

*Proof.* Take  $B$  of the moduli, from indices beyond any chosen point, and use Lemma 7.1 to find  $t$  far out with the  $i$ th of them dividing  $t + i$ . The interval  $[t, t + B)$  has length  $B$ , and  $u$  climbs by at most  $B$  per step while tending to infinity, so some  $u_s$  lands in it. Then  $u_s = t + i$  for some  $i < B$ , and the  $i$ th modulus divides  $u_s$ , against avoidance.  $\square$

Formalised as the [bounded-rise barrier](#). The key point is that the two hypotheses pull against each other exactly: divergence forces  $u$  to travel arbitrarily far, and bounded rise forbids it from stepping over a window of width  $B$ . Uniformity of the bound is what the proof uses, and it is the hypothesis that cannot be relaxed by this argument: a rise bounded only along a subsequence leaves the intervening steps free to jump the window. We have not looked for a sequence of unbounded rise avoiding every modulus. The statement is elementary and we are not aware of a prior published form of it, but we would not be surprised if one exists.

The barrier applies to the problem because a *reduced* exact tail already carries pairwise coprime moduli. Call  $(u, v)$  a reduced exact tail with multipliers  $a$  when  $\gcd(u_n, v_n) = 1$ ,

$$u_{n+1} + v_n = a_n u_n, \quad v_{n+1} = a_n v_n.$$

**Proposition 7.3** (reduced tails are pairwise coprime). *In a reduced exact tail,  $a_n$  is coprime to  $v_n$ ; the multipliers at distinct indices are pairwise coprime; and every earlier multiplier is coprime to every later numerator.*

These are the [step coprimality](#), the [pairwise coprimality](#), and the [whole-modulus avoidance](#). Feeding them to Theorem 7.2 gives the form used later: there is no reduced exact tail whose numerator tends to infinity with a uniformly bounded upward increment, the [reduced bounded-rise exclusion](#), together with its eventual form, the [eventual reduced exclusion](#).

The orbit of Section 2 is not reduced, so one more step is needed before the barrier can be applied to it: the common factor must stop changing. This is the only step connecting the orbit of Section 2 to the reduced tails of Proposition 7.3, and it is where the cofinal boundedness in the proof of Theorem 8.1 is used.

**Proposition 7.4** (the tail gcd stabilises). *Let  $(a, D, C)$  be an exact natural orbit whose centred state is negative cofinally, with magnitudes bounded along that cofinal set. Then  $\gcd(C_n, D_n)$  is eventually constant, and beyond that index the orbit divided by the stable gcd is a reduced exact tail.*

*Proof.* The tail gcd  $\gcd(C_n, D_n)$  divides its successor, the [gcd monotonicity](#), so the tail gcds form a positive divisibility chain. At an index where the centred state is negative the tail gcd is exactly  $\gcd(C_n, e_n)$ , the [gcd at a negative index](#), and is therefore at most the magnitude there; so the chain is bounded along the cofinal set of negative indices.

A positive divisibility chain whose values are bounded along a cofinal set is eventually constant, the [chain stabilisation](#). Hence cofinally bounded negative magnitudes make the tail gcd stabilise, the [gcd stabilisation](#), and beyond that point dividing by the stable gcd leaves the numerator and denominator coprime, that is, a reduced exact tail.  $\square$

The stabilisation of the tail gcd is the checked statement cited at the end of the proof; the last clause, that the divided orbit is a reduced exact tail, is exposition and carries no separate label. The hypothesis is boundedness *along a cofinal set* and not boundedness at every index. That is what the situation gives: the tail gcd is controlled only at the negative indices, and cofinally many of them is all the divisibility chain needs.

## 8 Excluding a bounded negative part

Assembling the previous section gives Theorem 8.1, the endpoint of the barrier argument. Theorem 9.1 in the next section is a second endpoint under a different finiteness hypothesis, and neither theorem contains the other. Theorem 9.1 assumes no orbit structure, only the state recurrence with  $C_n > 0$ , so it is not a special case of Theorem 8.1; conversely hypothesis (5) below does not imply finite normalised negative mass, since a constant negative part  $-b$  gives  $C_n \sim bn$  and hence a divergent  $\sum_n (-E_n)_+ / C_n$ . The hypotheses of Theorem 8.1 are listed in full, because two of them are analytic inputs that are assumed here rather than derived from the growth condition of Problem 1.1.

**Theorem 8.1** (bounded negative part). *Let  $a, C, D : \mathbb{N} \rightarrow \mathbb{N}$  and  $E : \mathbb{N} \rightarrow \mathbb{Z}$  satisfy*

1.  $a_n > 1$  and  $C_n > 0$  for every  $n$ ;
2. the exact dynamics  $C_{n+1} + D_n = a_n C_n$  and  $D_{n+1} = a_n D_n$ ;
3.  $E_n = \text{ctr}(a_n, D_n, C_n)$  for every  $n$ ;
4. eventual strict centring:  $|E_n| < C_n$  for all large  $n$ ;
5. eventually bounded negative part:  $-B \leq E_n$  for all large  $n$ , for some  $B$ ;
6. normalised vanishing: for every  $K$  there is an  $N$  with  $K|E_n| < C_n$  for all  $n \geq N$ .

Then  $E_n = 0$  for all sufficiently large  $n$ .

*Proof.* Suppose not. By Theorem 3.4 and its contrapositive form,  $E$  is nowhere zero beyond the centring threshold. If  $E$  is negative cofinally, then (5) bounds those magnitudes, so Proposition 7.4 makes the tail gcd stabilise and the orbit may be taken reduced past that point. Hypothesis (5) with Proposition 2.1 gives  $C_{n+1} \leq C_n + B$ , a bounded rise; hypothesis (6) with  $E$  nowhere zero gives  $C_n \rightarrow \infty$ . A reduced exact tail with a divergent numerator and bounded rise is excluded by Theorem 7.2. If instead  $E$  is eventually positive, the descent of Theorem 4.1 applies and forces  $E$  to vanish, contradicting nowhere-vanishing.  $\square$

Formalised as the [bounded-negative-part rigidity](#), with the eventual form, in which the centring and the bound need only hold from some index, as the [eventual bounded-negative-part rigidity](#). The two auxiliary results it uses are the [divergence from normalised vanishing](#) and the [exclusion of cofinally bounded negative magnitudes](#), the latter over the [tail-divergence form](#).

**Corollary 8.2.** *Under the hypotheses of Theorem 8.1, together with  $C_{n+1} \neq 0$  for all large  $n$ , the multipliers satisfy  $a_{n+1} = a_n^2 - a_n + 1$  for all sufficiently large  $n$ .*

*Proof.* Theorem 8.1 then Theorem 3.3. □

Hypotheses (1)–(3) are the state system. Hypothesis (4) says the centred state is a strict centred representative, and (6) says  $|E_n|/C_n \rightarrow 0$  in a division-free form. Neither is derived here from  $a_{n+1}/a_n^2 \rightarrow 1$ , and no argument in the formal source connects them to Problem 1.1. Theorem 8.1 is therefore a conditional theorem about the state system, and Corollary 8.2 does not settle any case of Problem 1.1. What is unconditional in this note is the exclusion of constant and periodic negative magnitudes, Theorems 5.1 and 6.1, and the barrier itself, Theorem 7.2.

## 9 Excluding finite normalised negative mass

Theorem 8.1 uses a uniform bound on the negative part. A different finiteness hypothesis, on the total normalised negative mass, removes a further family of orbits, and the argument is multiplicative rather than arithmetic: it compares  $C$  with a convergent product instead of with a system of congruences.

**Theorem 9.1** (finite normalised negative mass). *Let  $C_n \in \mathbb{N}_{>0}$  and  $E_n \in \mathbb{Z}$  satisfy*

$$C_{n+1} = C_n - E_n, \quad \forall K \exists N \forall n \geq N, \quad K|E_n| < C_n.$$

*If*

$$\sum_{n=0}^{\infty} \frac{(-E_n)_+}{C_n} < \infty,$$

*then  $E_n = 0$  eventually. For an exact reciprocal-tail orbit this implies  $a_{n+1} = a_n^2 - a_n + 1$  eventually.*

*Proof.* Put  $\delta_n = (-E_n)_+/C_n$ . The update gives  $C_{n+1} \leq C_n(1 + \delta_n)$ , so

$$C_N \leq C_0 \prod_{n < N} (1 + \delta_n).$$

The product is bounded because  $\sum \delta_n$  converges. Choose an integer  $M$  above that bound and apply normalised vanishing with  $K = M$ . Then  $M|E_n| < C_n < M$  for all sufficiently large  $n$ , forcing the integer  $E_n$  to be zero. The final recurrence is Theorem 3.3, whose hypothesis  $C_{n+1} \neq 0$  holds because  $C_n > 0$  throughout. □

The product argument is the [summable relative growth](#); its specialisation to negative mass is the [vanishing from finite negative mass](#), and the statement giving the recurrence directly is the [Sylvester recurrence from summable negative mass](#). This is a checked implication about the state system, not a derivation of normalised vanishing from Problem #243. Normalised vanishing is still an input: it is what converts the bound on  $C$  into a bound on  $|E_n|$ , and nothing here derives it from the growth condition of Problem 1.1.

## 10 Complements and further questions

Problem #243 is open. The negative case has narrowed, and it is worth being precise about what is left.

By Theorem 3.4 a counterexample has  $E_n \neq 0$  for every large  $n$ . By Theorem 4.1 it cannot have  $E$  eventually nonnegative. By Theorems 5.1 and 6.1 its negative magnitudes are neither eventually constant nor eventually periodic. By Theorem 8.1, if the analytic hypotheses (4) and (6) hold, the negative part cannot be bounded either, and by Theorem 9.1, under normalised vanishing, its normalised negative mass cannot be finite. What remains is the following.

**Problem 10.1** (unbounded divergent-mass negative excursions). Exclude, or construct, an exact orbit whose centred state is negative infinitely often with magnitudes unbounded along that cofinal set and

$$\sum_n \frac{(-E_n)_+}{C_n} = \infty.$$

Every argument in Sections 5–8 uses some finiteness hypothesis: a fixed set of prime divisors, a fixed period, or a fixed bound on the negative part. Theorem 9.1 uses a finite total multiplicative growth as well. An excursion of the kind described has none of these.

**Problem 10.2** (summable normalised negative mass). Prove from the rapid-growth and rationality hypotheses of Problem 1.1 that  $\sum_n (-E_n)_+/C_n < \infty$ , or prove any uniform bound on the partial products  $\prod_{n < N} (1 + (-E_n)_+/C_n)$ ; a bound of the form

$$\sum_{n=N}^M \frac{(-E_n)_+}{C_n} \leq B \quad \text{for all } M \geq N$$

with explicit hypotheses would suffice. A disproof should exhibit a coherent exact state orbit satisfying normalised vanishing while its negative mass diverges; a list of locally admissible defects is not enough, since local admissibility at each index does not produce a global orbit.

**Problem 10.3** (the two analytic hypotheses). Derive eventual strict centring and normalised vanishing from  $a_{n+1}/a_n^2 \rightarrow 1$  and rationality, or show that they do not follow. Until one or the other is done, Theorem 8.1 constrains the state system and not the original sequence.

**Problem 10.4** (the two published criteria). Formalise the criterion of Erdős and Straus stated in Section 1, and Duverney’s conditional characterisation, under explicit analytic hypotheses. Both are cited here and neither is formalised; each is stronger than every statement in this note.

### Statements and declarations

**Artefact and data availability.** The [pinned formal-source revision](#) contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

**Declaration of generative AI use.** Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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## A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision 580ae3a8bcce. The state system, the exclusions, the barrier, and the bounded-negative-part theorem are in `ReciprocalTailRigidity.lean`; the finite-negative-mass theorem is in `SparseResetRecovery.lean`; and the residue reduction of Appendix B is in `FiniteHorizonResidue.lean`. Three distinctions are worth carrying into the source. The identification of any of these modules with reciprocal tails is the exposition of Section 2 and is not a checked statement. The periodic exclusion assumes the regime  $e_n < a_n$ . And the final theorem assumes strict centring and normalised vanishing, which are hypotheses of the theorem and not consequences of the problem.

## B A factorial residue reduction for forced orbits

In the normalised constant-negative case the next multiplier is determined: at index  $n$  its numerator is

$$\text{num}(n, a) = (n + 1)a^2 - (n + 2)a + (n + 3),$$

the [forced numerator](#), and the divisor is  $n + 2$ . The orbit survives a step when that division is exact, giving a survival predicate, the [survival predicate](#). Deciding survival by iteration is expensive because the orbit grows doubly exponentially, and it is unnecessary: survival over a finite horizon depends on the initial value only through a factorial residue.

**Theorem B.1** (factorial residue reduction). *For all  $h$  and all integers  $a \equiv b \pmod{(h + 1)!}$ , the orbit from  $a$  survives  $h$  forced updates exactly when the orbit from  $b$  does.*

*Proof.* Let  $M(0, i) = 1$  and  $M(h + 1, i) = (i + 2)M(h, i + 1)$ , an ascending factorial with  $M(h, 0) = (h + 1)!$ . The numerator is a polynomial with integer coefficients, so congruences transfer; reducing the modulus gives divisibility by  $i + 2$  for one exactly when for the other, and cancelling that common factor from both values and modulus leaves the inductive hypothesis at  $M(h, i + 1)$ .  $\square$

Formalised as the [factorial residue reduction](#), over the [shrinking-modulus transport](#), the [polynomial congruence](#), and the [exact-division cancellation](#); the modulus identifications are the [ascending-factorial form](#) and the [factorial value at the initial index](#).

The search this supported is superseded. Running it over initial states below 5000 produced forced prefixes of length 17 and no longer, which was evidence and not a proof; Theorem 5.1 now excludes the constant-negative case outright, for every seed and at every scale. The reduction is retained because it is exact, and because the shrinking-modulus technique transfers to any forced orbit whose step is a polynomial division.

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